L A B O R A T O R Y 1

Orientation, Set-up & Project

environment, tooling and the semester project

Mobile and Embedded Computing · Dinu-Ștefan Rusu

Who's teaching this course

Dinu-Ștefan Rusu

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Microsoft Teams course channel

rusudinu.com

45+

Flutter apps shipped

400k

installs across them

Appeared in, spoken at & partnered with

What you'll build today

A working toolchain	Flutter and Dart installed, with flutter doctor reporting no blocking issues
A device to run on	An emulator or simulator that boots, with the default Flutter app running on it
A team and an idea	Teams of at most three, an idea written down, ready to be approved next lab
The hand-in workflow	Branch, pull request, repository: how every lab is submitted this semester

This lab puts Lecture 1 to work on the machine you will use all semester: the landscape, Git and GitHub.

What the laboratory is worth

3 p	1.5 p	1.5 p
Laboratory & assignments	**Lab test**	**Project**
individual, every session	individual, practical	teams of at most 3

4.5 points of laboratory plus 1.5 points of project. Six of the ten points of this course are earned in this room; the remaining four are the exam and the concept presentation.

THE SEMESTER PROJECT

Worth 1.5 points, in teams of at most three

One repository per team. You propose the idea, and we approve it in Laboratory 2

How the project runs, and how it is graded

Project development must be tracked on GitHub: one repository per team, from the first commit to the demo.

The application must be written in Flutter.

Individual participation is measured from your commits and from the quality of the code you wrote.

Every member must know their own part in detail, and have at least a brief awareness of the rest.

Teams of at most 3 students. You will present from your own laptop, at the projector.

WAY OF WORKING: ALSO GRADED

Pull requests, not pushes to main

A branch per feature

Code review before merge

GitHub Issues for the work items

Individual participation is measured from the commit history, not only from the final state of the code.

THE PROJECT

What your project must do

Five requirements. A project without all five is not approved.

State management, BLoC	every screen's state flows through a BLoC	Lecture 4
Real authentication	Firebase Auth or Keycloak, not a hard-coded login	Lecture 7
Persistence, offline-first	a real database; usable with no network	Lecture 10
Data follows the user	log in on another device and the data is there	Lecture 10
Realtime over WebSockets	at least one screen that updates live	Lecture 8

THE PROJECT

Forming a team and getting the idea approved

Teams of at most 3 students. Agree on an idea first, then write the team down.

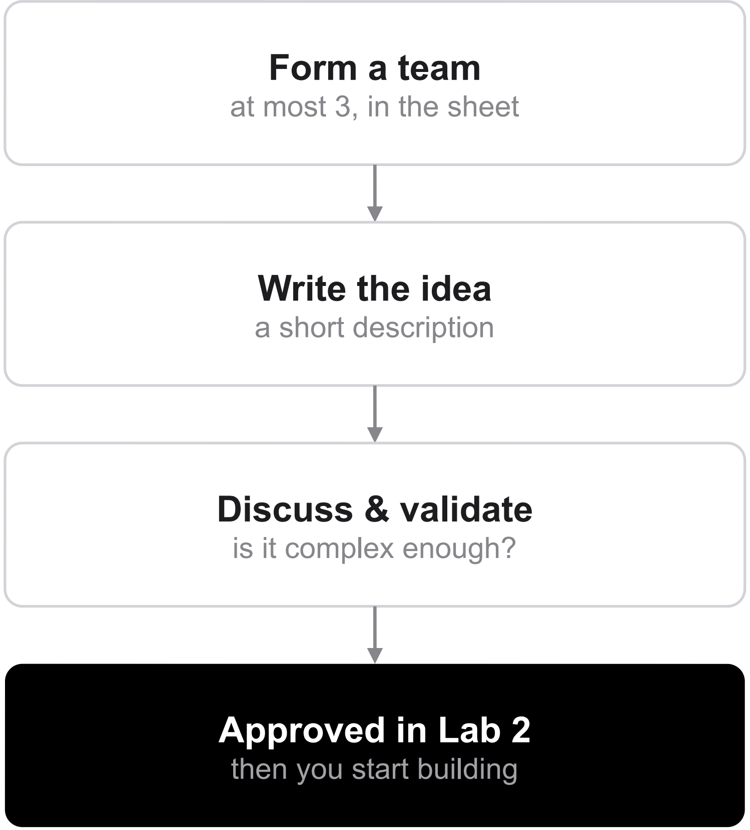
The team-forming sheet is posted in the Microsoft Teams channel of this course.

One row per team: the members, and a short description of what the app will do.

We then discuss the idea together and validate its complexity: aim for medium difficulty, with enough surface to carry all five requirements.

Approval happens in Laboratory 2. Do not start building before that conversation.

Nobody is assigned a project. You propose it, so propose something you actually want to build.



What you hand in at the end of the semester

The README must contain these sections, in this order

1. Team composition: one person per row with name, surname and group
2. Project description: what the app does, what screens it has
3. Link to the GitHub repository: if it is private, invite rusudinu as a collaborator
4. Other information: anything else you would like to mention

THE FINAL PRESENTATION

A live demo with slides, 15 minutes per team at the most

Slides go in the repository, in a folder named presentation, both .pptx and .pdf

You present from your own laptop, at the projector

Questions are asked about the code

Do not copy projects, and do not ship code you cannot explain. Every member is questioned on their own commits.

Everything you need to install

Flutter SDK	`https://docs.flutter.dev/install`	pick your OS, then Install manually
Android Studio	`https://developer.android.com/studio`	ships the Android SDK and the emulator
Visual Studio Code	`https://code.visualstudio.com`	add the Flutter extension
IntelliJ IDEA	`https://www.jetbrains.com/idea/download`	add the Flutter and Dart plugins
Xcode (macOS only)	`Mac App Store`	needed for the iOS simulator
Course Teams channel	`posted on Microsoft Teams`	the team-forming sheet lives here

Type the URLs exactly as printed; they are complete, on one line. The Teams channel carries the sheet, the announcements and every link that cannot be public.

TASK I

Set up your development environment

Work on your own laptop. This is the machine you will use all semester, including at the demo.

1. Install the Flutter SDK for your operating system from https://docs.flutter.dev/install
2. In the Install Flutter section of that page, follow the Install manually path
3. Add the flutter/bin folder to your PATH, so flutter --version works in a brand-new terminal
4. Install Android Studio from https://developer.android.com/studio
5. On macOS, also install Xcode from the Mac App Store
6. Install your editor, VS Code or IntelliJ IDEA, and add the Flutter and Dart plugins to it
7. Run flutter doctor and resolve everything it flags

Hints

flutter --version

flutter doctor · flutter doctor -v

PATH lives in ~/.zshrc on macOS, and in Environment Variables on Windows

Target: Flutter 3.44 with Dart 3.12

Done when

flutter --version prints a version in a fresh terminal

flutter doctor shows no blocking issue for Flutter, the Android toolchain or your editor

TASK II

Run the default app on a device

Straight after Task I, on the same machine. This is the deliverable for today.

1. Create an Android Virtual Device in Android Studio (Device Manager), or open the iOS Simulator on macOS
2. Start it and wait until it reaches the home screen
3. Run flutter devices and confirm your emulator appears in the list
4. Create a project with flutter create lab1_app
5. Run it with flutter run from inside that folder
6. Change the app-bar title in lib/main.dart, then press r to hot-reload and watch it update
7. Take a screenshot of the running app on your emulator

Hints

flutter devices

flutter emulators --launch <emulator_id>

flutter create lab1_app · flutter run

r hot-reloads, R hot-restarts, q quits

Done when

The counter app runs and the + button increments it

Your edited title appears after a hot reload

The screenshot is committed under /screenshots

When flutter doctor reports problems

```
# 1 - find out what is actually missing
flutter doctor -v

# 2 - accept the Android SDK licenses
flutter doctor --android-licenses

# 3 - SDK missing? Android Studio >
#     Settings > SDK Manager > SDK Tools >
#     Android SDK Command-line Tools

# 4 - check the fix landed
flutter doctor
```

cmdline-tools component is missing

Install Android SDK Command-line Tools from the SDK Manager, then run doctor again.

Android license status unknown

Run the licenses command above and answer y to every prompt.

Unable to locate Android SDK

Open Android Studio once; the SDK is downloaded on first launch.

No devices, emulators that will not boot, macOS

```
# nothing under connected devices?
flutter devices
flutter emulators
flutter emulators --launch <emulator_id>

# a physical Android phone works too:
# Developer options > USB debugging

# macOS - toolchain for the simulator
sudo xcode-select --install
sudo gem install cocoapods
```

The emulator never boots

Hardware acceleration is off. Turn on virtualization (VT-x / AMD-V) in the BIOS; on Windows use a WHPX image.

Apple silicon Mac

Pick an arm64 system image in the AVD Manager; x86 images will not run.

Xcode installed, still no simulator

Open Xcode once, accept the license, then install the command-line tools above.

If you are stuck for more than fifteen minutes, ask.

SUBMISSION

How to hand this in

Everything goes to your GitHub repository. No Word documents, no email, no Teams upload.

Commit your work to a branch named lab1, open a pull request, and merge it once it works.

Screenshots asked for in a task go in a /screenshots folder in the same repository.

Your commit history is part of the assessment. Commit as you go, not once at the end.

Checklist

code committed and pushed

screenshots where asked

app builds and runs

pull request merged

For Laboratory 1 the deliverable is the running default app: commit the project from Task II, with the screenshot in /screenshots.

Your team and project idea go in the team-forming sheet on Teams, not on GitHub.

Wrapping up

Recap

Flutter, an editor and the Android or iOS tooling installed

An emulator that boots, and the default app running on it

A team of at most three, and a project idea

Before Laboratory 2

Write your team and idea in the sheet on Teams

Come with a project you can defend for a semester

Bring the same laptop, in working order

Read more

docs.flutter.dev/install

docs.flutter.dev/get-started/test-drive

developer.android.com/studio/run

dart.dev/language